Critical Ultradifferentiable Geometry of the Fabius–Rvachev System

Local Thue–Morse universality, exact Denjoy–Carleman scales, discrete Legendre duality, Lambert–W saddles, and Bell-edge frontiers

Research report prepared for the ProveIt Fabius-function corpus

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Abstract

The exact derivative scale of the bounded Fabius function F_{bd} and the Rvachev up-function up is

$$W_n = 2^{n(n+1)/2}.$$

The repository develops this scale in dyadic jet identities, endpoint asymptotics, Fourier decay, compositional-iterate proofs, q-series, polynomial representations, and several other directions. The present report isolates a regularity-theoretic layer that is not developed there in interval-uniform form.

First, normalized derivatives $W_n^{-1} f^{(n)}$ obey a quantitative universal law on *every* nondegenerate interval, for $f = F_{\mathrm{bd}}$ or $f = \mathrm{up}$. Their pushforward measures converge in total variation, at rate $O(2^{-n}/|J|)$, to the symmetric law of $S\,\mathrm{up}(Y)$, where S is a Rademacher sign and Y is uniform on $[-1, 1]$. Consequently every sufficiently high derivative attains both sharp extrema $\pm W_n$ inside every interval; all local L^p norms and signed moments have universal limits.

Second, with

$$M_n = \frac{W_n}{n!} = \frac{2^{n(n+1)/2}}{n!},$$

we obtain an exact local Denjoy–Carleman classification. For any comparison weight N, membership of F_{bd} or up in the Roumieu class $\mathcal{E}^{\{N\}}$ is equivalent to exponential domination of M by N, while Beurling membership is equivalent to strict exponential domination. Thus M is the critical local weight on every interval. The class is nonquasianalytic and strongly nonquasianalytic, is stable under differentiation, composition, inversion, ordinary differential equations, and reciprocals, but fails moderate growth drastically. Its Faà di Bruno transform is computed exactly:

$$M_n^{\circ} = 2^n M_n.$$

Combining this calculus with the repository's dominant-spine theorem yields a new synthesis: every positive forward Fabius iterate and every positive inverse iterate belongs locally to $\mathcal{E}^{\{M\}}$ but to no $\mathcal{E}^{(M)}$ on any interval.

Third, optimizing the integration-by-parts Fourier bounds gives an exact discrete Legendre transform, including a one-periodic lattice correction. Comparison with the sharp sinc-product peak ray identifies the exact extra algebraic decay $\delta = \log_2(2/\sqrt{3})$. The factorial-normalized associated function has a different dual geometry: its saddle is governed by the lower real branch of Lambert W. These formulas extend naturally to geometric-base or q-deformed weights. Finally, exact Bell-polynomial computations through order 120 expose a new two-edge asymptotic hierarchy and motivate a precise $4/3$ limit conjecture. All proved, synthesized, numerical, and conjectural assertions are explicitly separated.

Status convention. A theorem labelled "proved here" is derived in this manuscript from explicitly stated baseline identities. A "proved synthesis" combines a theorem proved here with a theorem already established in the cited ProveIt document corpus. Numerical observations are not promoted to theorems. Conjectures and problems are marked as such. None of the new arguments is claimed to be Lean-verified.

Contents

1 Corpus audit, novelty boundary, and principal results

1.1 How the repository was screened

The live documentation tree contains a primary exposition, a mathematical glossary, a non-elementarity paper, a Lean walkthrough, a paper library, a canonical frontier volume, a large thematic draft collection, and a frozen historical archive [5, 9]. The draft manifest describes itself as a global inventory of every research draft and records the thematic relocations, provenance, and overlap assessments [8]. The archive in turn records that byte-identical historical copies were pruned when the live tree already preserved them [9].

For the present novelty screen, the two canonical LaTeX volumes were read as the main theorem and formula atlases, the manifest was used to traverse every active and historical draft family, and the closest-overlap LaTeX reports were read directly. The closest existing report is the forward-iterate manuscript [10]; it contains a pointwise Denjoy–Carleman interpretation but explicitly stops short of an interval-uniform classification. The separate Fourier corpus already proves the sharp logarithmic-Gaussian decay and peak-ray obstruction, so the present Fourier contribution is not a new decay theorem: it is the exact Carleman duality and optimal comparison with the derivative envelope.

This is a corpus-wide semantic audit, not a claim of absolute publication priority. The word “new” below means “not found in the screened ProveIt LaTeX corpus in this form.” Frozen or absorbed duplicate documents were not treated as independent sources of mathematical novelty.

1.2 Overlap map

Existing layer	Closest corpus location	Boundary respected in this report
Dyadic derivatives, Thue–Morse signs, exact cells, moments	Primary Fabius–Rvachev exposition [6]	Used as baseline. The new layer is a quantitative *local probability law* on every interval, with total-variation rate and exact local extrema.
Endpoint asymptotics and Lambert phases	Canonical frontier volume and Lambert guide [7, 8]	Endpoint transseries are not repeated. Lambert W appears instead as the saddle of the factorial-normalized critical weight.
Sharp Fourier decay of the infinite sinc product	Dedicated eight-document Fourier corpus and audits [8]	The sharp exponent is imported. We derive the exact integer optimization, periodic rounding factor, and the precise algebraic gap between Fourier decay and derivative regularity.
Faà di Bruno defects and self-iterates	Forward-iterate report [10]	Its cancellation-sensitive spine theorem is not reproved. The new object is the factorial-normalized weight transform M°, computed exactly, and its use in an interval-class synthesis for all forward and inverse iterates.
q-Pochhammer, Gaussian coefficients, inverse q-analogs	q-series monograph and several inverse/expansion reports [8]	No branch-inversion theory is duplicated. We give a base-b/q dictionary for the critical Denjoy–Carleman weight, its associated function, and its lattice dual.

Existing layer	Closest corpus location	Boundary respected in this report
Legendre, Lagrange, Bell–Bernoulli, Stein–Koopman, total positivity	Representation and probability compilations [8]	These representations are not reproduced. Bell polynomials enter only as the positive majorant naturally attached to the newly identified critical weight.

1.3 Principal contributions

The main claims are as follows.

R1. **proved here**: a quantitative local Thue–Morse universality theorem in total variation, with error at most $6\,2^{1-n}/|J|$.

R2. **proved here**: exact sharp extrema $\pm W_n$ on every fixed interval for all large n, and universal local L^p and moment limits.

R3. **proved here**: an if-and-only-if local Roumieu/Beurling classification against *every* positive comparison sequence N.

R4. **proved here**: exact weight geometry, including strong nonquasianalyticity, failure of moderate growth, and the closed formula $M_n^\circ = 2^n M_n$.

R5. **proved synthesis**: every positive forward and inverse compositional iterate has critical local class $\mathcal{E}^{\{M\}} \setminus \mathcal{E}^{(M)}$ and quadratic order $1/2$.

R6. **proved here**: an exact discrete Legendre formula for the Fourier–Carleman envelope, with a bounded one-periodic phase correction.

R7. **proved synthesis**: the sinc-product arithmetic contributes exactly $\delta = \log_2(2/\sqrt{3})$ powers beyond the derivative envelope, and no larger uniform power is possible.

R8. **proved here**: the standard associated function of M has a W_{-1}-controlled saddle and an explicit logarithmic expansion.

R9. **proved here**: the weight calculus extends to geometric base $b \geq 2$, or $q = b^{-1}$, including an exact base-b Faà di Bruno transform.

R10. **conjectural**: a Bell-edge expansion whose first unremoved term is $\frac{4}{3}n^6 4^{-n}$; exact integer arithmetic through $n = 120$ supports it.

2 Baseline identities and notation

2.1 Fabius, Rvachev, and Thue–Morse conventions

Let $F_{\mathrm{bd}} : [0,1] \to [0,1]$ be the bounded Fabius function and let $\mathrm{up} : \mathbb{R} \to \mathbb{R}$ be the compactly supported Rvachev up-function, normalized by

$$\operatorname{supp}\mathrm{up} = [-1,1], \qquad \mathrm{up}(0) = 1, \qquad \int_{-1}^{1} \mathrm{up}(y)\,\mathrm{d}y = 1.$$

On the unit interval,

$$F_{\mathrm{bd}}(x) = \mathrm{up}(x-1), \qquad 0 \leq x \leq 1. \tag{1}$$

The signed Thue–Morse sequence is

$$\varepsilon_k = (-1)^{\mathrm{wt}_2(k)}, \tag{2}$$

where $\mathrm{wt}_2(k)$ is the sum of the binary digits of k.

Write

$$T_n = \frac{n(n+1)}{2}, \qquad W_n = 2^{T_n}, \qquad h_n = 2^{1-n}. \tag{3}$$

The exact cell map is

$$c_{n,k}(y) = -1 + \frac{2k+1+y}{2^n}, \qquad -1 \le y \le 1, \qquad 0 \le k < 2^n. \tag{4}$$

It maps $[-1, 1]$ affinely onto the kth cell of length h_n in $[-1, 1]$. The primary exposition proves the exact derivative cell identity

$$\mathrm{up}^{(n)}\big(c_{n,k}(y)\big) = W_n \varepsilon_k \, \mathrm{up}(y). \tag{5}$$

In particular,

$$\left\|\mathrm{up}^{(n)}\right\|_{L^\infty([-1,1])} = W_n, \qquad \mathrm{up}^{(n)}\big(c_{n,k}(0)\big) = W_n \varepsilon_k. \tag{6}$$

The shift (1) transfers these identities to F_{bd} on $[0, 1]$. Equivalent matched-grid identities and the global signed extension of F_{bd} are recorded in [6, 10].

2.2 A discrepancy lemma

The local theorem rests on a particularly strong elementary feature of Thue–Morse: every interval of indices has uniformly bounded signed discrepancy.

Lemma 2.1 (Prefix and block discrepancy). *Let $P(N) = \sum_{k=0}^{N-1} \varepsilon_k$. Then*

$$P(2m) = 0, \qquad P(2m+1) = \varepsilon_m. \tag{7}$$

Consequently, for all integers $0 \le a < b$,

$$\left| \sum_{k=a}^{b-1} \varepsilon_k \right| \le 2. \tag{8}$$

Moreover, no three consecutive signs are equal.

Proof. The identities $\varepsilon_{2m} = \varepsilon_m$ and $\varepsilon_{2m+1} = -\varepsilon_m$ give (7) by pairing. Hence every prefix sum belongs to $\{-1, 0, 1\}$, and a block sum is the difference of two prefix sums. Finally, each pair $(\varepsilon_{2m}, \varepsilon_{2m+1})$ consists of opposite signs, so a run of length three is impossible. □

2.3 Fourier convention

We use

$$\widehat{f}_{2\pi}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi i \xi t} \, \mathrm{d}t. \tag{9}$$

For $\Phi = \widehat{\mathrm{up}}_{2\pi}$, the repository's product formula is

$$\Phi(\xi) = \prod_{j=0}^{\infty} \mathrm{sinc}_{\mathrm{rad}}\left(\frac{\pi \xi}{2^j}\right), \qquad \mathrm{sinc}_{\mathrm{rad}}(z) = \frac{\sin z}{z}. \tag{10}$$

The exact cell formula also gives

$$\left\|\mathrm{up}^{(n)}\right\|_{L^1(\mathbb{R})} = W_n. \tag{11}$$

Indeed, on every cell the absolute value is $W_n \, \mathrm{up}(y)$ and $\mathrm{d}x = 2^{-n} \, \mathrm{d}y$; summing the 2^n cells uses $\int \mathrm{up} = 1$.

3 A universal local law for high derivatives

The global derivative distribution follows immediately from the exact cells. The new point is that every fixed interval, however short and wherever placed, sees the same limiting law, with an explicit total-variation error.

3.1 The limiting probability measure

Let Y be uniform on $[-1,1]$ and let S be independent with $\mathbb{P}(S=1)=\mathbb{P}(S=-1)=1/2$. Define the probability measure

$$\mu_* = \operatorname{Law}(S\,\operatorname{up}(Y)). \tag{12}$$

Equivalently, for every bounded Borel function H,

$$\int H(z)\,\mathrm{d}\mu_*(z) = \frac{1}{4}\int_{-1}^{1}\big(H(\operatorname{up}(y)) + H(-\operatorname{up}(y))\big)\,\mathrm{d}y. \tag{13}$$

For a nondegenerate interval J in the domain of f, where $f \in \{F_{\mathrm{bd}}, \operatorname{up}\}$, let $\mu^f_{n,J}$ be the pushforward of normalized Lebesgue measure on J by $W_n^{-1} f^{(n)}$.

Theorem 3.1 (Quantitative local Thue–Morse universality). *Let $f = \operatorname{up}$ with $J \subseteq [-1,1]$, or $f = F_{\mathrm{bd}}$ with $J \subseteq [0,1]$. Assume $|J|>0$. Then for every bounded Borel H,*

$$\left|\frac{1}{|J|}\int_J H\left(\frac{f^{(n)}(x)}{W_n}\right)\mathrm{d}x - \int H\,\mathrm{d}\mu_*\right| \le 6\,\|H\|_\infty \frac{h_n}{|J|}. \tag{14}$$

Consequently,

$$\operatorname{TV}\left(\mu^f_{n,J} - \mu_*\right) \le 6\frac{h_n}{|J|} = \frac{12}{2^n|J|}, \tag{15}$$

where $\operatorname{TV}(\nu) = \sup_{\|H\|_\infty \le 1}\left|\int H\,\mathrm{d}\nu\right|$.

Proof. It suffices to prove the assertion for up, because F_{bd} is its translate on $[0,1]$. Decompose J into the union of all complete level-n cells contained in J and at most two boundary fragments. Let q be the number of complete cells, let their consecutive index set be K, and let $r = |J| - qh_n$. Then $0 \le r \le 2h_n$.

Put

$$A_+ = \int_{-1}^{1} H(\operatorname{up}(y))\,\mathrm{d}y, \qquad A_- = \int_{-1}^{1} H(-\operatorname{up}(y))\,\mathrm{d}y.$$

By (5), the contribution of the complete cells is

$$I_K = \frac{h_n}{2}\sum_{k\in K} A_{\varepsilon_k}.$$

Since $qh_n\int H\,\mathrm{d}\mu_* = qh_n(A_+ + A_-)/4$, their difference is

$$I_K - qh_n\int H\,\mathrm{d}\mu_* = \frac{h_n}{4}\left(\sum_{k\in K}\varepsilon_k\right)(A_+ - A_-).$$

By Lemma 2.1, its absolute value is at most $2h_n\|H\|_\infty$.

The integral on the boundary fragments has absolute value at most $r\|H\|_\infty$, while $r\left|\int H\,\mathrm{d}\mu_*\right| \le r\|H\|_\infty$. Thus replacing the fragments by their limiting mass costs at most $2r\|H\|_\infty \le 4h_n\|H\|_\infty$. Adding the two errors and dividing by $|J|$ proves (14). Taking the supremum over $\|H\|_\infty \le 1$ proves (15). ☐

Remark 3.2 (Strength of the convergence). Weak convergence would follow from continuous test functions. Here the estimate is uniform over all bounded Borel tests, so discontinuous threshold events and local quantiles converge at the same dyadic rate. The only sources of error are the two cut cells and the bounded Thue–Morse sign imbalance.

3.2 Exact local extrema and oscillation

Corollary 3.3 (Every interval eventually contains both sharp signs)**.** *Under the hypotheses of Theorem 3.1, if*

$$h_n \le \frac{|J|}{5}, \tag{16}$$

then J contains at least three complete level-n cells and hence cells of both Thue–Morse signs. Therefore

$$\max_{x\in J} f^{(n)}(x) = W_n, \qquad \min_{x\in J} f^{(n)}(x) = -W_n, \tag{17}$$

and

$$\left\| f^{(n)} \right\|_{L^\infty(J)} = W_n, \qquad \operatorname{osc}_J f^{(n)} = 2W_n. \tag{18}$$

In particular these equalities hold for every sufficiently large n on every fixed nondegenerate interval.

Proof. At most two cells meeting J fail to be complete, so the number q of complete cells satisfies $q \ge |J|/h_n - 2 \ge 3$. A consecutive block of at least three Thue–Morse signs contains both signs by Lemma 2.1. At the midpoint of a positive or negative cell, (5) and $\mathrm{up}(0) = 1$ give the values $\pm W_n$. The global bound (6) shows that they are the extrema. ☐

This result is much more rigid than a global supremum statement. It says that flatness at the endpoints does not lower the derivative class on any one-sided neighborhood: arbitrarily high derivatives recover both global extrema at distances of order 2^{-n}.

3.3 Universal local moments and L^p norms

Corollary 3.4 (Local L^p asymptotics)**.** *For every $p > 0$,*

$$\lim_{n\to\infty} \frac{1}{|J|W_n^p} \int_J \left| f^{(n)}(x) \right|^p \mathrm{d}x = \frac{1}{2} \int_{-1}^{1} \mathrm{up}(y)^p \,\mathrm{d}y. \tag{19}$$

For $p = 1$ this becomes

$$\left\| f^{(n)} \right\|_{L^1(J)} = \frac{|J|}{2} W_n + O(h_n W_n). \tag{20}$$

More generally, for every integer $m \ge 0$,

$$\lim_{n\to\infty} \frac{1}{|J|W_n^m} \int_J f^{(n)}(x)^m \,\mathrm{d}x = \begin{cases} 0, & m \textit{ odd}, \\ \dfrac{1}{2} \displaystyle\int_{-1}^{1} \mathrm{up}(y)^m \,\mathrm{d}y, & m \textit{ even}. \end{cases} \tag{21}$$

All errors are $O(h_n/|J|)$ after normalization, with constants depending only on the bounded test function.

Proof. Apply Theorem 3.1 to $H(z) = |z|^p$ and $H(z) = z^m$ on $[-1, 1]$. For $p = 1$, use $\int_{-1}^{1} \mathrm{up} = 1$. ☐

Corollary 3.5 (Exact local quadratic derivative order)**.** *For $f = F_{\mathrm{bd}}$ or $f = \mathrm{up}$ and every nondegenerate interval J in its domain,*

$$\lim_{n\to\infty} \frac{\log_2 \left\| f^{(n)} \right\|_{L^\infty(J)}}{n^2} = \frac{1}{2}. \tag{22}$$

Proof. By Corollary 3.3, the numerator equals $\log_2 W_n = n(n+1)/2$ for all sufficiently large n. ☐

4 The exact local Denjoy–Carleman scale

4.1 Roumieu and Beurling conventions

Let $N = (N_n)_{n\geq 0}$ be any positive sequence. On a compact interval J, write $f \in \mathcal{E}^{\{N\}}(J)$ when there exist $C, \rho > 0$ such that

$$\left\| f^{(n)} \right\|_{L^\infty(J)} \leq C\rho^n n! N_n \qquad (n \geq 0). \tag{23}$$

Write $f \in \mathcal{E}^{(N)}(J)$ when for every $\rho > 0$ there exists $C_\rho > 0$ for which (23) holds. These are the standard factorial-normalized Denjoy–Carleman conventions [4, 2].

Define the critical sequence

$$M_n = \frac{W_n}{n!} = \frac{2^{n(n+1)/2}}{n!}, \qquad M_0 = 1. \tag{24}$$

4.2 Complete comparison theorem

Theorem 4.1 (Exact local comparison against every weight). *Let $f = F_{\mathrm{bd}}$ or $f = \mathrm{up}$, and let J be any nondegenerate compact interval in its domain. For every positive sequence N,*

$$f \in \mathcal{E}^{\{N\}}(J) \iff \limsup_{n\to\infty} \left(\frac{M_n}{N_n}\right)^{1/n} < \infty, \tag{25}$$

$$f \in \mathcal{E}^{(N)}(J) \iff \limsup_{n\to\infty} \left(\frac{M_n}{N_n}\right)^{1/n} = 0. \tag{26}$$

Thus the local regularity class is independent of the position and length of J; only the finite threshold after which the sharp derivative values occur depends on J.

Proof. By Corollary 3.3,

$$\left\| f^{(n)} \right\|_{L^\infty(J)} = W_n = n! M_n$$

for every sufficiently large n. If (23) holds, division by $n!N_n$ and passage to nth roots gives the necessity of (25). Conversely, finite limsup means that $M_n \leq C\rho^n N_n$ for some C, ρ after increasing C to absorb finitely many initial terms.

The Beurling statement is the same argument with every $\rho > 0$. A positive sequence a_n satisfies $a_n \leq C_\rho \rho^n$ for every $\rho > 0$ if and only if $\limsup a_n^{1/n} = 0$. □

Corollary 4.2 (Critical Roumieu but not critical Beurling). *On every nondegenerate interval J,*

$$f \in \mathcal{E}^{\{M\}}(J) \quad \textit{but} \quad f \notin \mathcal{E}^{(M)}(J). \tag{27}$$

The weight M is therefore exact up to ordinary exponential equivalence.

Proof. In Theorem 4.1 take $N = M$. The comparison ratio is identically one. □

Corollary 4.3 (No finite Gevrey order on any interval). *For every finite $s \geq 1$, neither F_{bd} nor up belongs locally on any nondegenerate interval to the Gevrey class with derivative bounds $C\rho^n (n!)^s$.*

Proof. The Gevrey-s sequence is $N_n = (n!)^{s-1}$. Stirling's formula gives

$$\frac{1}{n} \log\left(\frac{M_n}{N_n}\right) = \frac{\log 2}{2} n - s\log n + O(1) \longrightarrow +\infty.$$

Thus even the Roumieu criterion fails. □

Corollary 4.4 (Sharp transition among quadratic exponential classes). *For $\alpha > 0$, let $\mathcal{Q}_\alpha$ denote the class with derivative bound $C\rho^n 2^{\alpha n^2}$. On every nondegenerate interval,*

$$f \in \mathcal{Q}_\alpha^{\{\cdot\}} \iff \alpha \geq \frac{1}{2}, \tag{28}$$

$$f \in \mathcal{Q}_\alpha^{(\cdot)} \iff \alpha > \frac{1}{2}. \tag{29}$$

At $\alpha = 1/2$, the residual factor $2^{n/2}$ is exactly of Roumieu exponential type but not of Beurling type.

Proof. Use $N_n = 2^{\alpha n^2}/n!$. Then

$$\left(\frac{M_n}{N_n}\right)^{1/n} = 2^{(1/2-\alpha)n+1/2}.$$

Apply Theorem 4.1. □

4.3 A local regularity spectrum

The theorem can be packaged as an intrinsic spectrum. For any smooth g on a compact interval J, define

$$\tau_J(g; M) = \limsup_{n\to\infty} \left(\frac{\|g^{(n)}\|_{L^\infty(J)}}{n!M_n}\right)^{1/n}. \tag{30}$$

Then $g \in \mathcal{E}^{\{M\}}(J)$ exactly when $\tau_J(g; M) < \infty$, and $g \in \mathcal{E}^{(M)}(J)$ exactly when $\tau_J(g; M) = 0$. For the basic functions,

$$\tau_J(F_{\text{bd}}; M) = \tau_J(\text{up}; M) = 1. \tag{31}$$

Thus every interval has the same exact Carleman type, even intervals adjacent to a flat endpoint.

5 Geometry of the critical weight

The critical sequence is unusual: it has enough convexity for a full nonlinear calculus and enough growth to be nonquasianalytic, yet it is far outside the moderate-growth regime.

5.1 Quotients, nonquasianalyticity, and derivations

Theorem 5.1 (Exact structural properties of M). *The sequence* (24) *has the following properties.*

(i) Its quotients are

$$\mu_n := \frac{M_n}{M_{n-1}} = \frac{2^n}{n} \qquad (n \geq 1). \tag{32}$$

Hence M is logarithmically convex and $M_n^{1/n}$ is increasing.

(ii) It is nonquasianalytic, with exact Denjoy–Carleman sum

$$\sum_{k=0}^{\infty} \frac{M_k}{(k+1)M_{k+1}} = \sum_{k=0}^{\infty} 2^{-(k+1)} = 1. \tag{33}$$

(iii) It is strongly nonquasianalytic. For $k \geq 1$,

$$\sum_{j\geq k} \frac{M_{j-1}}{jM_j} = 2^{1-k} \leq 2\frac{M_{k-1}}{M_k}. \tag{34}$$

(iv) It is derivation closed:

$$\sup_{n\geq 1}\left(\frac{M_{n+1}}{M_n}\right)^{1/n} < \infty, \qquad \lim_{n\to\infty}\left(\frac{M_{n+1}}{M_n}\right)^{1/n} = 2. \tag{35}$$

(v) It does not have moderate growth. In fact

$$\frac{M_{2n}}{M_n^2} = \frac{2^{n^2}}{\binom{2n}{n}} \geq 2^{n^2-2n}. \tag{36}$$

Proof. The quotient formula is immediate from $T_n - T_{n-1} = n$. Moreover, $\mu_{n+1}/\mu_n = 2n/(n+1) \geq 1$, proving log-convexity. The remaining exact sums follow from

$$\frac{M_k}{(k+1)M_{k+1}} = 2^{-(k+1)}, \qquad \frac{M_{j-1}}{jM_j} = 2^{-j}.$$

The derivation condition follows from $M_{n+1}/M_n = 2^{n+1}/(n+1)$. Finally,

$$\frac{M_{2n}}{M_n^2} = 2^{T_{2n}-2T_n}\frac{(n!)^2}{(2n)!} = \frac{2^{n^2}}{\binom{2n}{n}},$$

and $\binom{2n}{n} \leq 4^n$. □

Remark 5.2. Nonquasianalyticity is consistent with the compact support of up. Moderate growth, by contrast, would require a uniform exponential control of $M_{j+k}/(M_jM_k)$; the quadratic exponent creates an unavoidable factor 2^{jk}, so moderate growth fails by an exponentially widening margin.

5.2 The exact Faà di Bruno transform

For a positive sequence M, define the Faà di Bruno transform

$$M_n^\circ = \max\left\{M_jM_{\alpha_1}\cdots M_{\alpha_j} : j\geq 1,\ \alpha_i \geq 1,\ \sum_{i=1}^{j}\alpha_i = n\right\}, \qquad M_0^\circ = 1. \tag{37}$$

This is the normalized weight that controls all chain-rule partitions [2].

Theorem 5.3 (Exact Faà di Bruno envelope). *For the critical sequence $M_n = 2^{T_n}/n!$,*

$$M_n^\circ = 2^nM_n \qquad (n\geq 0). \tag{38}$$

For $n \geq 1$, the lower bound is attained by the partition $j = n$, $\alpha_1 = \cdots = \alpha_n = 1$.

Proof. The indicated partition gives $M_nM_1^n = 2^nM_n$, because $M_1 = 2$.

For the upper bound, fix j and a composition $\alpha_1 + \cdots + \alpha_j = n$. Log-convexity implies that, for fixed sum, the product $\prod_i M_{\alpha_i}$ is maximized when one part is as large as possible and the others equal 1. Put

$$r = n - j + 1.$$

It suffices to estimate $M_jM_rM_1^{j-1}$. Direct calculation gives

$$\frac{M_jM_rM_1^{j-1}}{M_n} = 2^{j-(j-1)(r-1)}\frac{n!}{j!r!}. \tag{39}$$

Since $n = j + r - 1$,

$$\frac{n!}{j!r!} \leq \binom{j+r-1}{r-1} \leq (j+1)^{r-1} \leq 2^{j(r-1)}. \tag{40}$$

The middle inequality follows, for example, by encoding weakly increasing choices by words on $j+1$ symbols, and the last uses $j+1 \le 2^j$. Substitution into (39) gives

$$\frac{M_j M_r M_1^{j-1}}{M_n} \le 2^{j-(j-1)(r-1)+j(r-1)} = 2^{j+r-1} = 2^n.$$

Taking the maximum proves the result. □

Remark 5.4 (Distinction from the iterate defect). The iterate report [10] studies the *signed, unnormalized* Faà di Bruno expansion of $F_{\mathrm{bd}}^{\circ r}$ and proves that branched partitions are small after cancellation-sensitive normalization by W_n. The present theorem answers a different question: it computes the *positive, factorial-normalized* sequence transform that governs closure of the entire Denjoy–Carleman class. The two results are complementary.

5.3 Nonlinear stability

The hypotheses in the Rainer–Schindl stability theorem are now explicit: $M_n^{1/n} \to \infty$, derivation closure holds, and $M_n^\circ \le 2^n M_n$. Therefore both types of the critical class possess the full standard nonlinear calculus.

Corollary 5.5 (Critical classes are stable under nonlinear operations)**.** *The Roumieu and Beurling classes $\mathcal{E}^{\{M\}}$ and $\mathcal{E}^{(M)}$ are stable under composition, local inversion at points with nonzero Jacobian, solution of ordinary differential equations, and reciprocal of nonvanishing functions.*

Proof. Apply the Roumieu and Beurling equivalence theorems of Rainer and Schindl [2] using Theorems 5.1 and 5.3. □

The coexistence of Equation (36) and Corollary 5.5 is important: moderate growth is sufficient for many convenient tensor-product and extension results, but it is not necessary for the one-dimensional nonlinear operations needed by the Fabius dynamics.

6 Inverse Fabius function and all compositional iterates

Let $Q_F : (0,1) \to (0,1)$ be the inverse of F_{bd}, and write

$$G_r = F_{\mathrm{bd}}^{\circ r}, \qquad H_r = Q_F^{\circ r} \qquad (r \ge 1).$$

6.1 The inverse function at the critical scale

Theorem 6.1 (Exact interior class of the inverse Fabius function)**.** *On every compact interval $K \Subset (0,1)$,*

$$Q_F \in \mathcal{E}^{\{M\}}(K) \quad \textit{and} \quad Q_F \notin \mathcal{E}^{(M)}(K). \tag{41}$$

Consequently the inverse Fabius function is not of finite Gevrey order on any nondegenerate interior interval.

Proof. The image $J = Q_F(K)$ is compactly contained in $(0,1)$, where $F'_{\mathrm{bd}} > 0$. By Corollaries 4.2 and 5.5, local inversion gives $Q_F \in \mathcal{E}^{\{M\}}(K)$. If $Q_F \in \mathcal{E}^{(M)}(K)$, Beurling inversion stability would imply $F_{\mathrm{bd}} \in \mathcal{E}^{(M)}(J)$, contradicting Corollary 4.2. The Gevrey conclusion follows as in Corollary 4.3. □

This theorem concerns interior regularity. It does not replace the repository's much finer endpoint asymptotics for $Q_F(y)$ as $y \downarrow 0$, where Lambert phases and log-periodic transseries enter. Rather, it identifies the unique ambient ultradifferentiable scale through which those endpoint singular asymptotics must connect to the smooth interior.

6.2 Forward and inverse iterates: an exact synthesis

The forward-iterate report proves the following dominant-spine lower bound. For each fixed $r \geq 1$, there is a co-countable dense set $E_r \subset (0,1)$ such that, for every $x \in E_r$, infinitely many n satisfy

$$\left|G_r^{(n)}(x)\right| \geq c(x,r) W_n \mathcal{H}_r(x)^n, \qquad c(x,r) > 0, \quad \mathcal{H}_r(x) > 0. \tag{42}$$

See [10]. That report uses this to prove zero Taylor radius and notes that it is a pointwise rather than interval-uniform Denjoy–Carleman statement. The exact class calculus above closes that gap.

Theorem 6.2 (Critical regularity of every positive iterate)**.** *Fix $r \geq 1$.*

(i) On every nondegenerate compact interval $J \subseteq [0,1]$,

$$G_r \in \mathcal{E}^{\{M\}}(J) \quad \textit{but} \quad G_r \notin \mathcal{E}^{(M)}(J). \tag{43}$$

(ii) On every compact interval $K \Subset (0,1)$,

$$H_r \in \mathcal{E}^{\{M\}}(K) \quad \textit{but} \quad H_r \notin \mathcal{E}^{(M)}(K). \tag{44}$$

(iii) Their local quadratic derivative order is exactly $1/2$ in limsup:

$$\limsup_{n\to\infty} \frac{\log_2 \left\|G_r^{(n)}\right\|_{L^\infty(J)}}{n^2} = \frac{1}{2}, \qquad \limsup_{n\to\infty} \frac{\log_2 \left\|H_r^{(n)}\right\|_{L^\infty(K)}}{n^2} = \frac{1}{2}. \tag{45}$$

Statements (i)–(iii) are a proved synthesis*: the upper bounds use the new critical-class calculus, while the lower obstruction uses the repository's spine theorem.*

Proof. Because $F_{\mathrm{bd}} \in \mathcal{E}^{\{M\}}$ and the class is composition stable, $G_r \in \mathcal{E}^{\{M\}}$ locally. Suppose $G_r \in \mathcal{E}^{(M)}(J)$ for a nondegenerate interval J. Choose $x \in E_r \cap \operatorname{int} J$. For every $\rho > 0$ there would be C_ρ such that

$$\left|G_r^{(n)}(x)\right| \leq C_\rho \rho^n W_n.$$

Taking $\rho < \mathcal{H}_r(x)$ contradicts (42) along its infinite subsequence. This proves (i).

The inverse iterate H_r is locally Roumieu by repeated inversion and composition. If it were Beurling on K, inversion stability would place its inverse G_r in the Beurling class on $H_r(K)$, contradicting (i). This proves (ii).

Roumieu membership gives an upper bound $\|g^{(n)}\| \leq C\rho^n W_n$, so the limsup in (45) is at most $1/2$. For G_r, the pointwise lower bound gives the reverse inequality. For H_r, if the limsup were strictly smaller than $1/2$, then $\|H_r^{(n)}\|/W_n$ would decay faster than every ordinary exponential and hence H_r would be Beurling, contradicting (ii). ☐

Corollary 6.3 (No iterate belongs locally to any finite Gevrey class)**.** *Every positive forward or inverse iterate is outside every finite-order Gevrey class on every nondegenerate interval in its natural domain.*

Proof. Every finite Gevrey class embeds in $\mathcal{E}^{(M)}$ because $(n!)^s/W_n$ decays superexponentially. Apply Theorem 6.2. ☐

6.3 A sharper type formula suggested by the spine expansion

The theorem determines the quadratic order and the Roumieu/Beurling boundary, but not the exact ordinary-exponential type

$$\tau_J(G_r; M) = \limsup_n \left(\frac{\left\|G_r^{(n)}\right\|_{L^\infty(J)}}{W_n} \right)^{1/n}.$$

The spine theorem identifies the pointwise candidate

$$\mathcal{H}_r(x) = \max_{0\leq j<r} (F_{\mathrm{bd}}^{\circ j})'(x).$$

Conjecture 6.4 (Uniform spine-type formula). *For every $r \geq 1$ and every compact nondegenerate $J \Subset (0,1)$,*

$$\tau_J(G_r; M) = \max_{x\in J} \mathcal{H}_r(x). \tag{46}$$

An analogous formula holds for inverse iterates with the corresponding inverse-orbit spine weights.

The lower inequality follows from the dense pointwise spine theorem and continuity of $\mathcal{H}_r$. The open part is a uniform upper estimate through tie points and dyadic orbit preimages. It should be approachable by making the weighted-defect remainder uniform on compact intervals and controlling the finite dominant-spine sum without discarding its orbit geometry.

7 Fourier–Carleman duality and its lattice correction

7.1 Optimization of all integration-by-parts bounds

Using (11), n integrations by parts give

$$|\Phi(\xi)| \leq \frac{W_n}{(2\pi|\xi|)^n} = \frac{2^{n(n-1)/2}}{(\pi|\xi|)^n}. \tag{47}$$

Define the optimized derivative envelope

$$B_C(\xi) = \inf_{n\in\mathbb{N}_0} \frac{2^{n(n-1)/2}}{(\pi|\xi|)^n}. \tag{48}$$

Theorem 7.1 (Exact discrete Legendre transform). *For $|\xi| \geq 1/(\pi\sqrt{2})$, set*

$$a(\xi) = \log_2(\pi|\xi|) + \frac{1}{2}, \qquad d(\xi) = \operatorname{dist}(a(\xi), \mathbb{Z}). \tag{49}$$

Then every nearest integer to $a(\xi)$ is an optimizer in (48), and

$$B_C(\xi) = 2^{-a(\xi)^2/2 + d(\xi)^2/2}. \tag{50}$$

Equivalently, with

$$E_\kappa(x) = \exp\left(-\frac{(\log x)^2}{2\log 2}\right) x^{-\kappa}, \qquad \kappa_C = \frac{1}{2} + \log_2 \pi, \tag{51}$$

we have

$$B_C(\xi) = C_C E_{\kappa_C}(|\xi|) P(a(\xi)), \tag{52}$$

where

$$C_C = \exp\left(-\frac{(\log\pi + \frac{1}{2}\log 2)^2}{2\log 2}\right), \tag{53}$$

$$P(a) = 2^{\operatorname{dist}(a,\mathbb{Z})^2/2}, \qquad 1 \leq P(a) \leq 2^{1/8}. \tag{54}$$

Thus the exact envelope is a logarithmic Gaussian times an algebraic factor and a bounded one-periodic lattice wave in $\log_2|\xi|$.

Proof. Taking base-2 logarithms of the nth term in (48) gives

$$\frac{n(n-1)}{2} - n\log_2(\pi|\xi|) = \frac{(n-a)^2 - a^2}{2}.$$

The minimum over integers is therefore obtained at a nearest integer to a, and its value is (50). Expanding a^2 in powers of $\log|\xi|$ yields (52)–(53). Finally $0 \leq \operatorname{dist}(a,\mathbb{Z}) \leq 1/2$ gives (54). ☐

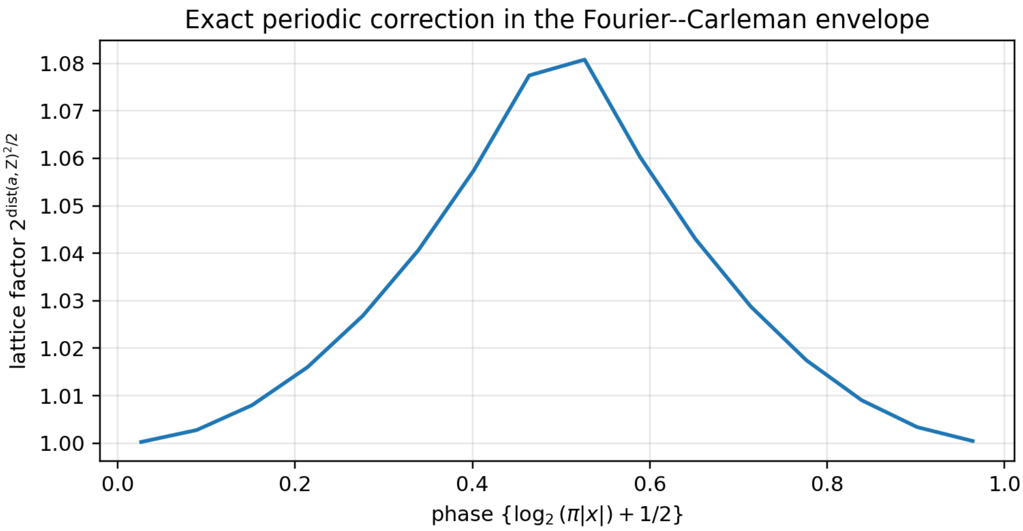


Figure 1: The exact lattice factor in (54). It is not a numerical artifact: it is the rounding cost in the discrete Legendre transform.

7.2 What the derivative scale does and does not explain

The repository's sharp Fourier theorem gives

$$|\Phi(\xi)| \le CE_{\kappa_\infty}(|\xi|), \qquad \kappa_\infty = \frac{3}{2} + \log_2\left(\frac{\pi}{\sqrt{3}}\right), \tag{55}$$

and proves that no larger algebraic exponent can replace κ_∞ globally; the obstruction occurs on the peak ray

$$\xi_k = 2^k \frac{2}{3}. \tag{56}$$

The exact derivative envelope has exponent κ_C. Their difference is

$$\delta = \kappa_\infty - \kappa_C = 1 - \frac{1}{2}\log_2 3 = \log_2\left(\frac{2}{\sqrt{3}}\right) \approx 0.2075187496394219. \tag{57}$$

Theorem 7.2 (Exact algebraic gain beyond derivative regularity)**.** *There exists $C > 0$ such that, for $|\xi| \ge 1$,*

$$|\Phi(\xi)| \le CB_C(\xi)|\xi|^{-\delta}. \tag{58}$$

The exponent δ is optimal: for every $\eta > 0$ there is no constant C_η with

$$|\Phi(\xi)| \le C_\eta B_C(\xi)|\xi|^{-\delta-\eta} \qquad (|\xi| \ge 1). \tag{59}$$

Indeed, along (56),

$$\frac{|\Phi(\xi_k)|}{B_C(\xi_k)\xi_k^{-\delta}} = R_* > 0 \qquad (k \ge 0), \tag{60}$$

with numerically $R_* \approx 0.6484307237021603$.

Proof. Since the periodic factor P in (52) is bounded above and below, (55) is equivalent, up to a constant, to (58).

For the peak ray, the sinc product factors exactly as

$$\left|\Phi(2^k \tfrac{2}{3})\right| = \left|\Phi(\tfrac{2}{3})\right| \left(\frac{3\sqrt{3}}{\pi}\right)^k 2^{-k(k-1)/2-3k}. \tag{61}$$

Moreover $a(\xi_k) = k + a(2/3)$, so the distance $d(\xi_k)$ to the integer lattice is independent of k. Substitution of (57) into (50) shows that the powers of 2 and the linear powers of ξ_k match exactly, proving (60). A bound with $\delta + \eta$ would force the left side of (60) times ξ_k^{η} to remain bounded, which is impossible. □

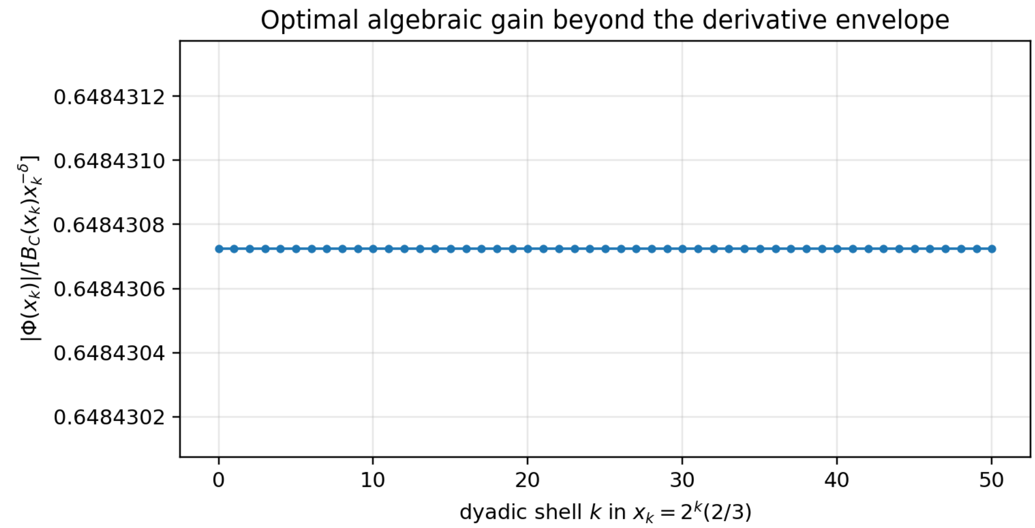


Figure 2: The normalized peak-ray ratio in (60). Its phase-locked constancy confirms the exact algebraic gap δ; the plotted variation is floating-point roundoff.

Remark 7.3 (Interpretation). The quadratic logarithmic Gaussian is already forced by the derivative scale W_n. The infinite sinc product contributes additional arithmetic information: the repeated numerator $|\sin(2^m \cdot 2\pi/3)| = \sqrt{3}/2$ along the peak ray yields exactly δ further powers. Thus ultradifferentiable regularity determines the carrier, while lacunary trigonometric arithmetic determines the sharp algebraic refinement.

8 The associated function and a Lambert–W_{-1} saddle

The Fourier envelope above optimizes the *raw* derivative scale after the 2π integration-by-parts factor has cancelled one power of 2 per derivative. The standard Denjoy–Carleman associated function instead uses the factorial-normalized sequence M:

$$\omega_M(t) = \sup_{n\geq 0} \log\left(\frac{t^n}{M_n}\right) = \sup_{n\geq 0}\left(n\log t + \log(n!) - \frac{\log 2}{2} n(n+1)\right). \tag{62}$$

The factorial introduces an entropy term $n\log n$, and the optimizer is no longer an affine function of $\log t$.

8.1 Continuous saddle and Lambert predictor

Put $L = \log 2$ and extend the objective continuously:

$$G_t(s) = s\log t + \log\Gamma(s+1) - \frac{L}{2}s(s+1), \qquad s \geq 0. \tag{63}$$

Its large saddle s_t satisfies

$$\log t + \psi(s_t+1) - L\left(s_t + \frac{1}{2}\right) = 0, \tag{64}$$

where $\psi = \Gamma'/\Gamma$. Ignoring the $O(1/s)$ difference between $\psi(s+1)$ and $\log s$ gives the explicit Lambert predictor

$$\lambda(t) = -\frac{1}{L} W_{-1}\left(-\frac{\sqrt{2}L}{t}\right). \tag{65}$$

The lower real branch is forced because the relevant saddle tends to infinity.

Theorem 8.1 (Lambert saddle and associated-function expansion). *As $t \to \infty$,*

$$s_t = \lambda(t) + O\left(\frac{1}{\log t}\right), \tag{66}$$

$$n_*(t) = s_t + O(1), \tag{67}$$

where $n_*(t)$ is any integer optimizer in (62). *At the continuous saddle,*

$$G_t(s_t) = \frac{L}{2}s_t^2 - s_t + \frac{1}{2}\log(2\pi s_t) - \frac{1}{2} + O\left(\frac{1}{s_t}\right). \tag{68}$$

Consequently, with $u = \log t$,

$$\omega_M(t) = \frac{u^2}{2L} + \frac{u\log u}{L} - \left(\frac{1}{2} + \frac{1+\log L}{L}\right)u + O\big((\log u)^2\big). \tag{69}$$

Proof. The expansion $\psi(s+1) = \log s + 1/(2s) + O(s^{-2})$ transforms (64) into

$$Ls - \log s = \log t - \frac{L}{2} + O(1/s).$$

The equation without the error is solved by (65). Its left derivative is $L - 1/s$, bounded away from zero for large s, so the saddle perturbation is $O(1/s) = O(1/\log t)$. Strict concavity for large s gives (67).

At a saddle, substitute $\log t = L(s+1/2) - \psi(s+1)$ into (63) to obtain

$$G_t(s) = \frac{L}{2}s^2 - s\psi(s+1) + \log\Gamma(s+1).$$

Stirling expansions of $\log\Gamma$ and ψ yield (68). Finally, the Lambert equation gives

$$Ls_t = u + \log u - \log L - \frac{L}{2} + O\left(\frac{\log u}{u}\right).$$

Substitution into (68), together with the $O(1)$ rounding loss from (67), yields (69). □

8.2 Two dualities, two periodic phenomena

There are now two distinct optimization problems.

(i) The Fourier integration-by-parts envelope minimizes a pure quadratic over the integer lattice. Its optimizer is exactly a rounded affine function of $\log_2|\xi|$, and its correction is the explicit periodic factor $P(a)$.

(ii) The standard associated function maximizes a quadratic plus $\log\Gamma(n+1)$. The entropy term shifts the saddle by $\log\log t$ and introduces Lambert W_{-1}. Integer rounding still produces a bounded phase, but now around a nonlinear Lambert trajectory.

This distinction prevents a common conflation: the Lambert correction is not needed to optimize the raw Fourier derivative bound, while the exact lattice wave of Theorem 7.1 is invisible in a purely continuous associated-function asymptotic.

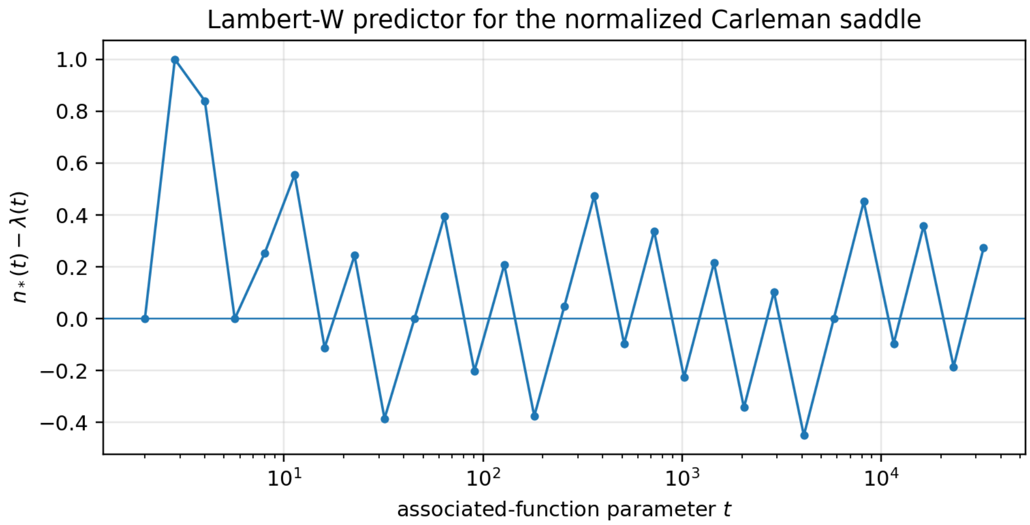


Figure 3: The exact discrete optimizer $n_*(t)$ minus the Lambert predictor (65). Integer rounding produces the bounded sawtooth; the continuous saddle error is much smaller.

9 Geometric-base and q-deformed critical weights

The corpus contains geometric-base atomic sinc splines and extensive q-analog machinery. The regularity scale suggested by those constructions is

$$W_n^{(b)} = b^{n(n+1)/2}, \qquad M_n^{(b)} = \frac{b^{n(n+1)/2}}{n!}, \qquad b > 1. \tag{70}$$

Writing $q = b^{-1}$ makes this $M_n^{(q)} = q^{-n(n+1)/2}/n!$, the same quadratic exponent that pervades Gaussian q-coefficients.

Theorem 9.1 (Base-b weight calculus). *Let $b \geq 2$ and $L_b = \log b$. Then:*

(i)

$$\frac{M_n^{(b)}}{M_{n-1}^{(b)}} = \frac{b^n}{n}, \tag{71}$$

so $M^{(b)}$ is log-convex.

(ii)

$$\sum_{k=0}^{\infty} \frac{M_k^{(b)}}{(k+1)M_{k+1}^{(b)}} = \frac{1}{b-1}, \tag{72}$$

and

$$\sum_{j \geq k} \frac{M_{j-1}^{(b)}}{j M_j^{(b)}} = \frac{b^{1-k}}{b-1} \leq \frac{b}{b-1} \frac{M_{k-1}^{(b)}}{M_k^{(b)}}. \tag{73}$$

(iii) Moderate growth fails:

$$\frac{M_{2n}^{(b)}}{(M_n^{(b)})^2} = \frac{b^{n^2}}{\binom{2n}{n}}. \tag{74}$$

(iv) The Faà di Bruno transform is exact:

$$\left(M^{(b)}\right)_n^\circ = b^n M_n^{(b)}. \tag{75}$$

(v) *The associated-function saddle has predictor*

$$\lambda_b(t) = -\frac{1}{L_b} W_{-1}\left(-\frac{\sqrt{b}\, L_b}{t}\right). \tag{76}$$

(vi) *The pure quadratic lattice transform satisfies, for $A > 1$,*

$$\inf_{n\geq 0} \frac{b^{n(n-1)/2}}{A^n} = b^{-a_b^2/2+\mathrm{dist}(a_b,\mathbb{Z})^2/2}, \qquad a_b = \log_b A + \frac{1}{2}. \tag{77}$$

Proof. The quotient, sums, and moderate-growth ratio are direct calculations. For the Faà di Bruno transform, repeat the proof of Theorem 5.3. The factorial ratio is bounded by $2^{j(r-1)} \leq b^{j(r-1)}$, while the all-singleton partition gives the matching lower bound $(M_1^{(b)})^n = b^n$. The Lambert and lattice formulas repeat Theorems 7.1 and 8.1 with $\log 2$ replaced by $\log b$. □

Remark 9.2 (The range $1 < b < 2$). The quotient b^n/n is eventually increasing but may fail log-convexity at finitely many initial indices. A finite regularization leaves the associated Denjoy–Carleman class unchanged. The exact constant in (75) can then change at finitely many orders, while an exponential Faà di Bruno bound and the same asymptotic Lambert/lattice geometry remain.

Problem 9.3 (Local universality beyond base two)**.** *The binary proof of Theorem 3.1 uses two special facts: an exact cell profile and uniformly bounded discrepancy of the Thue–Morse signs. Determine which base-b atomic spline systems possess a finite-state sign or phase cocycle with bounded interval discrepancy. In such systems, prove a local pushforward law and identify whether its limit is discrete, absolutely continuous, or singular.*

10 Bell-polynomial edge asymptotics

10.1 The positive critical-norm majorant

The exact Faà di Bruno transform controls the largest normalized weight of a single chain-rule partition. A different object sums all positive Bell terms:

$$C_n = \frac{1}{W_n} \sum_{k=1}^{n} W_k B_{n,k}(W_1, W_2, \ldots, W_{n-k+1}), \tag{78}$$

where $B_{n,k}$ is the exponential partial Bell polynomial. This is the critical norm majorant one obtains by replacing every derivative amplitude in a self-composition by its sharp absolute bound and discarding cancellation. It is deliberately not identified with an actual derivative of $F_{\mathrm{bd}} \circ F_{\mathrm{bd}}$.

The two edges $k \approx n$ and $k \approx 1$ dominate, while the middle is suppressed by the quadratic weight defect. Several terms can be extracted exactly.

Proposition 10.1 (Explicit Bell edges)**.** *For $n \geq 4$, the following contributions to C_n are exact:*

$$\begin{aligned} k = n: &\quad 2^n, &(79)\\ k = n-1: &\quad n(n-1), &(80)\\ k = 1: &\quad 2, &(81)\\ k = n-2: &\quad 2^{3-n}\left(2\binom{n}{3} + 3\binom{n}{4}\right). &(82)\end{aligned}$$

The endpoint split $(1, n-1)$ inside the $k = 2$ term contributes exactly

$$16n\, 2^{-n}. \tag{83}$$

The partition with $k = n-3$ and three blocks of size 2 contributes exactly

$$960\binom{n}{6} 4^{-n} \sim \frac{4}{3} n^6 4^{-n}. \tag{84}$$

Proof. For $k = n$, $B_{n,n} = W_1^n$, giving 2^n. For $k = n-1$, there is one block of size 2 and $n-2$ singletons, so

$$\frac{W_{n-1}}{W_n}\binom{n}{2}W_2W_1^{n-2} = n(n-1).$$

The $k = 1$ term is $W_1W_n/W_n = 2$.

For $k = n-2$, the excess $n-k = 2$ is realized either by one 3-block or two 2-blocks. Their Bell multiplicities are $\binom{n}{3}$ and $3\binom{n}{4}$, and substitution of $W_j = 2^{T_j}$ gives (82).

For $k = 2$, the two endpoint compositions $(1, n-1)$ combine to nW_1W_{n-1} inside $B_{n,2}$, yielding (83) after multiplication by W_2/W_n.

Finally, three 2-blocks and $n-6$ singletons have multiplicity $15\binom{n}{6}$. Their normalized weight is

$$\frac{W_{n-3}W_2^3W_1^{n-6}}{W_n} = 2^{6-2n} = 64\,4^{-n},$$

which gives (84). □

10.2 A two-edge asymptotic conjecture

Define the explicitly subtracted approximation

$$\begin{aligned} A_n = {} & 2^n + n(n-1) + 2 \\ & + 2^{3-n}\left(2\binom{n}{3} + 3\binom{n}{4}\right) + 16n\,2^{-n}. \end{aligned} \tag{85}$$

All values in the accompanying computation are exact rational numbers until the final decimal output.

Conjecture 10.2 (First unremoved Bell edge). *As $n \to \infty$,*

$$C_n - A_n \sim \frac{4}{3}n^6 4^{-n}. \tag{86}$$

Equivalently,

$$\lim_{n\to\infty}\frac{4^n}{n^6}(C_n - A_n) = \frac{4}{3}. \tag{87}$$

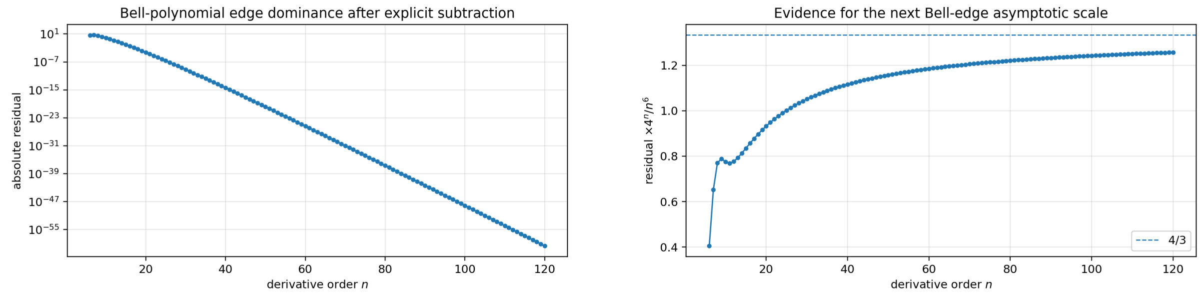
The candidate constant is not curve fitting: it is forced by the three-pair term (84). Other $k = n-3$ block patterns have lower polynomial degree, while the $k = 3$ left edge is of order $n^2 4^{-n}$. Terms farther from either edge carry an additional exponential defect. A proof should convert this heuristic into a uniform two-edge tail estimate.

Table 2: Exact-arithmetic evidence for Conjecture 10.2.

n	$4^n(C_n - A_n)/n^6$
20	0.932702130656337
40	1.116793440375664
60	1.185331851851859
80	1.220944228515625
100	1.242749164800000
120	1.257470214763374

Conjecture 10.3 (Full Bell-edge transseries). *The sequence C_n has an asymptotic expansion organized by distance from the two Faà di Bruno edges. For fixed right defect $d = n-k$, the leading term comes from d two-blocks and has the form*

$$\frac{(2d)!}{2^d d!}\binom{n}{2d}2^{d(d+1)/2-(d-1)n}. \tag{88}$$



(a) Unscaled residual. (b) Scaled residual and the conjectural $4/3$ limit.

Figure 4: Two-edge Bell-polynomial numerics through $n = 120$.

For fixed left order k, the leading term comes from one block of size $n - k + 1$ and $k - 1$ singletons. After subtracting finitely many terms from both edges, the remaining middle is exponentially smaller than the first omitted edge scale.

This conjecture would turn the positive Bell majorant into a precise asymptotic counterpart of the signed partition-defect theory in [10]. It may also supply a reusable template for base-b and q-Bell compositions.

11 Further conjectures and a research program

11.1 Microlocal and Fourier questions

Problem 11.1 (Fourier characterization at the critical weight). *Develop a Paley–Wiener or almost-holomorphic characterization tailored to $M_n = 2^{T_n}/n!$ that retains the exact lattice phase of Theorem 7.1. Standard equivalence up to exponential constants forgets the factor $P(a)$; the goal is a phase-sensitive refinement that recovers it from complex extension or short-time Fourier data.*

Conjecture 11.2 (A periodic ultradifferentiable wavefront profile). *For compactly supported functions saturating the critical derivative sequence, the optimal logarithmic-Gaussian Fourier carrier admits a bounded one-periodic profile in $\log_2 |\xi|$. For up, the derivative-lattice profile P and the sinc-product arithmetic profile combine multiplicatively, with the peak ray selecting a fixed phase.*

11.2 Inverse endpoints and two-regime regularity

Problem 11.3 (Interior critical class versus endpoint transseries). *Match the interior class $\mathcal{E}^{\{M\}} \setminus \mathcal{E}^{(M)}$ of $Q_F = F_{\mathrm{bd}}^{-1}$ to the known Lambert–periodic endpoint expansion as $y \downarrow 0$. A satisfactory connection should identify a scale-dependent crossover: derivatives on compact interior intervals are governed by M, while the shrinking endpoint neighborhood introduces the Lambert saddle and periodic endpoint phase.*

Conjecture 11.4 (Endpoint-rescaled Carleman type). *There exists a two-parameter asymptotic for $\sup_{y\in[\eta,2\eta]} |Q_F^{(n)}(y)|$ in the joint limit $n \to \infty$, $\eta \downarrow 0$, whose transition curve is expressed through the same W_{-1} phase that governs the endpoint inverse expansion. Away from that curve it reduces either to an M_n-type interior bound or to the endpoint transseries.*

11.3 Extension theory without moderate growth

The exact weight is strongly nonquasianalytic but not moderate. This places it between two familiar regimes: compactly supported functions and nonlinear calculus are available, while several standard Whitney-extension and tensor-product theorems cannot be invoked automatically.

Problem 11.5 (Optimal descendant and Whitney extension)**.** *Construct the optimal descendant weight associated with M and determine whether Whitney jets on intervals or tame closed sets admit extension without loss, with a finite loss, or only through a weight matrix. Express any loss in the Lambert associated-function scale* (69)*.*

11.4 q, cyclotomic, and arithmetic deformations

Problem 11.6 (Cyclotomic Carleman geometry)**.** *For complex q approaching a root of unity, combine the branch geometry of the corpus's inverse q-analogs with the quadratic weight $q^{-T_n}/n!$. Determine how the associated Lambert saddle bifurcates when* $\log q$ *becomes multivalued and whether the lattice correction becomes a higher-dimensional torus phase.*

Conjecture 11.7 (Base-deformed exact type)**.** *For every geometric atomic spline in the corpus whose derivatives have exact L^1 scale b^{T_n}, the sharp Fourier envelope decomposes into*

base-b quadratic carrier $\times$ integer-lattice phase $\times$ arithmetic gain.

The first two factors are universal consequences of the critical weight; the third is determined by the extremal orbit of the base-b trigonometric numerator.

11.5 Formalization roadmap

The following sequence is designed to minimize dependencies.

- **L1.** Formalize $P(2m) = 0$, $P(2m+1) = \varepsilon_m$, and block discrepancy at most 2.
- **L2.** Combine the existing exact cell identity with interval-cell decomposition to formalize Theorem 3.1 first for simple functions H, then for bounded Borel tests or, more modestly, continuous tests.
- **L3.** Formalize the eventual local extrema theorem, which needs only three complete cells and the no-three-equal-sign lemma.
- **L4.** Define interval Roumieu/Beurling seminorms and prove the sequence-comparison criterion Theorem 4.1.
- **L5.** Formalize the quotient identities, nonquasianalytic sums, and failure of moderate growth.
- **L6.** Define the Faà di Bruno sequence transform and prove $M_n^\circ = 2^n M_n$ using the one-large-part reduction.
- **L7.** Import the existing iterate spine theorem and derive the exact critical class of all forward and inverse iterates.
- **L8.** Formalize the integer quadratic optimizer and lattice factor in Theorem 7.1; this is independent of analytic Fourier machinery.
- **L9.** Formalize the Lambert predictor only after a reusable real W_{-1} asymptotic library is in place.
- **L10.** Treat the Bell-edge conjecture last; exact recurrences and finite certificates can be formalized before the asymptotic tail estimate.

12 Reproducible numerical experiments

No numerical computation is used to prove the local universality theorem, the exact Denjoy–Carleman classification, the weight geometry, the Faà di Bruno transform, the iterate synthesis, or the Fourier lattice formula. The script `frontier_experiments.py` has four narrower purposes:

- (i) verify the direct integer infimum and the closed form (50) over a logarithmic frequency grid;
- (ii) evaluate the exact peak-ray factorization and the normalized constant in (60);
- (iii) compare the exact integer optimizer of ω_M with the Lambert predictor;

(iv) compute C_n by exact integer Bell recurrences through $n = 120$ and test Conjecture 10.2.

The command is

```
python3 frontier_experiments.py --out-dir .
```

The dependencies are Python 3, mpmath, and matplotlib. CSV outputs preserve long decimal strings; Bell-polynomial arithmetic uses Python integers and fractions.Fraction until serialization. Both vector PDF and PNG figures are generated. The archive includes the script, requirements file, all CSV tables, figures, a numerical summary, and a SHA-256 ledger.

Table 3: Numerical constants independently reproduced by the script.

Quantity	Value
$\Phi(2/3)$	$0.32160188864654939804635794159529\,21\ldots$
κ_C	$2.15149612947231879804327929510800\,73\ldots$
κ_∞	$2.35901487911174070731640982313409\,91\ldots$
δ	$0.20751874963942190927313052802609\,17\ldots$
R_*	$0.64843072370216024975605409187029\,98\ldots$

13 Conclusions

The dyadic derivative factor $2^{n(n+1)/2}$ is more than a convenient upper bound. It is a complete local regularity invariant.

The exact cells and bounded Thue–Morse discrepancy force every interval to sample the same normalized derivative law. That local law upgrades a global sharp bound to exact local extrema, moment limits, and a complete comparison theorem against all Denjoy–Carleman sequences. Factorial normalization then reveals a critical weight that is simultaneously nonquasianalytic, nonmoderate, and closed under the nonlinear operations central to the Fabius system. Its exact Faà di Bruno transform makes the closure quantitative.

The same scale also clarifies two apparently different parts of the corpus. On the dynamical side, it converts the existing dominant-spine lower bounds into an exact Roumieu-versus-Beurling classification for every forward and inverse iterate. On the Fourier side, it supplies the logarithmic-Gaussian carrier, while the sinc arithmetic contributes an exactly measurable extra algebraic gain. The discrete integer optimizer leaves a periodic lattice fingerprint; the factorial-normalized associated function instead introduces Lambert W_{-1} through its entropy-shifted saddle.

This regularity layer is compatible with, but not a duplication of, the corpus's q-analog, Lambert, Fourier, inverse, Legendre, Bell–Bernoulli, and representation programs. It provides a common scale on which those programs can interact. The most immediate next targets are the uniform iterate type formula, a phase-sensitive Fourier characterization, a rigorous Bell-edge transseries, and a base-b local universality theorem.

A Additional proof details

A.1 Why the one-large-part product is maximal

For completeness, suppose M is log-convex, so its quotients $\mu_n = M_n/M_{n-1}$ are increasing. If $2 \le a \le b$, then

$$\frac{M_{a-1}M_{b+1}}{M_aM_b} = \frac{\mu_{b+1}}{\mu_a} \ge 1.$$

Thus transferring one unit from a smaller part a to a larger part b does not decrease the product. Iterating pushes every composition $\alpha_1 + \cdots + \alpha_j = n$ to $(n-j+1, 1, \ldots, 1)$ and proves the reduction used in Theorem 5.3.

A.2 Right-edge Bell exponents

Let $k = n - d$ and write the block sizes as $\alpha_i = 1 + \beta_i$, with $\beta_i \geq 0$ and $\sum_i \beta_i = d$. The normalized base-2 exponent of the corresponding Bell weight is

$$E(d, \boldsymbol{\beta}) = T_{n-d} + \sum_i T_{1+\beta_i} - T_n = -(d-1)n + \frac{d^2 + \sum_i \beta_i^2}{2}. \tag{89}$$

For fixed d, the largest polynomial multiplicity occurs when all positive β_i equal 1, i.e. when there are d two-blocks. There are

$$\frac{(2d)!}{2^d d!}\binom{n}{2d}$$

such set partitions, and their exponent is $d(d+1)/2 - (d-1)n$, giving (88). This calculation explains both the exact $d = 3$ candidate and the expected hierarchy in Conjecture 10.3.

A.3 The peak-ray constant

Let

$$a_0 = \log_2(2\pi/3) + \frac{1}{2}, \qquad d_0 = \operatorname{dist}(a_0, \mathbb{Z}).$$

Then (60) has the explicit value

$$R_* = |\Phi(2/3)| \left(\frac{2}{3}\right)^{\delta} 2^{a_0^2/2 - d_0^2/2}. \tag{90}$$

The equality for all k is exact; the numerical script evaluates $\Phi(2/3)$ by a high-precision tail-controlled sinc product.

B Artifact inventory and reproducibility

The accompanying archive has the following layout.

Path	Purpose
`fabius_carleman_frontiers.tex`	Complete LaTeX source.
`fabius_carleman_frontiers.pdf`	Rendered report.
`frontier_experiments.py`	Commented deterministic numerical experiment.
`requirements.txt`	Minimum Python dependencies.
`README.txt`	Build and reproduction instructions.
`numerical_summary.txt`	Key constants and run scope.
`data/*.csv`	Gauge, peak-ray, Lambert-saddle, and Bell data.
`figures/*.pdf`	Vector figures used by the report.
`figures/*.png`	Raster inspection copies.
`CORPUS_AUDIT.txt`	Scope and novelty-screen record.
`SHA256SUMS.txt`	Checksums for all archive payloads.

The PDF was built with `latexmk`, rendered page-by-page for visual inspection, and checked for structural and font issues. The numerical script is not network-dependent.

References

[1] J. Arias de Reyna, Arithmetic of the Fabius function, *Integers* **18** (2018), Article A51, 20 pp.; arXiv:1702.06487.

[2] A. Rainer and G. Schindl, Equivalence of stability properties for ultradifferentiable function classes, *Rev. R. Acad. Cienc. Exactas Fís. Nat. Ser. A Mat. RACSAM* **110** (2016), 17–32, doi:10.1007/s13398-014-0216-0.

[3] A. Rainer and G. Schindl, Composition in ultradifferentiable classes, *Studia Math.* **224** (2014), 97–131, doi:10.4064/sm224-2-1.

[4] A. Rainer, Arc-smooth functions on closed sets, *Compos. Math.* **155** (2019), 645–680, arXiv:1801.08335.

[5] V. Reshetnikov, *ProveIt: Analysis/FabiusFunction/docs*, live documentation tree, `https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs`, accessed 30 August 2026.

[6] V. Reshetnikov, *Fabius Function and Rvachev Up*, canonical LaTeX exposition, `https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex`, accessed 30 August 2026.

[7] V. Reshetnikov, *Semi-formalized Research Frontiers*, canonical LaTeX volume, `https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex`, accessed 30 August 2026.

[8] V. Reshetnikov, *Draft Manifest*, global thematic inventory and provenance map, `https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/MANIFEST.md`, accessed 30 August 2026.

[9] V. Reshetnikov, *Archive of Historical Document Versions*, `https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/archive`, accessed 30 August 2026.

[10] V. Reshetnikov, *Nowhere Analyticity of Every Positive Compositional Iterate of the Fabius Function: A Dyadic-Spine Proof through Faà di Bruno Partitions*, `https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/representations/fabius_iterates_nowhere_analytic/fabius_iterates_nowhere_analytic.tex`, 30 August 2026.